MP363 – Quantum Mechanics I

Problem Set 4

Due by the end of tutorial on Thursday, 4 December 2014

1. Suppose a system is in the N^{th} eigenstate of an infinite square well, namely, the wavefunction $\psi(t, x)$ is zero for $x \leq 0$ and $x \geq a$, and

$$\psi(t, x) = \sqrt{\frac{2}{a}} e^{-iE_N t/\hbar} \sin\left(\frac{N\pi x}{a}\right), \qquad E_N = \frac{\pi^2 \hbar^2}{2ma^2} N^2$$

for $0 < x < a$.

   (a) Show that the expectation value of X is $a/2$ and the expectation value of P is zero.

   (b) Find $\langle X^2 \rangle$ and $\langle P^2 \rangle$.

   (c) The uncertainty in the measurement of an observable A is

$$\Delta a \quad = \quad \sqrt{\langle A^2 \rangle - \langle A \rangle^2}.$$

   Using this and your results from (a) and (b), find expressions for Δx and Δp and use them to confirm that $\Delta x \Delta p$ is greater than $\hbar/2$ for all values of N.

2. Suppose the wavefunction for a particle in an infinite square well is given (for $0 < x < a$) by

$$\psi(x) \quad = \quad Cx(a - x)$$

where C is some constant.

   (a) Determine the value of C such that ψ is normalised.

   (b) Using

$$X\psi(x) = x\psi(x),\ P\psi(x) = -i\hbar \frac{\mathrm{d}\psi}{\mathrm{d}x}(x),\ H\psi(x) = -\frac{\hbar^2}{2m} \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}(x),$$

   find the expectation values $\langle X \rangle$, $\langle X^2 \rangle$, $\langle P \rangle$, $\langle P^2 \rangle$ and $\langle H \rangle$.

   (c) Use your results from (b) to confirm that $\Delta x \Delta p$ is larger than $\hbar/2$ for this state.

   (d) Find the probability that a measurement of energy gives $E_2 = 2\pi^2\hbar^2/ma^2$.